



Employee Salary Prediction Using Machine Learning



Project Overview

Employee Salary Prediction is a Machine Learning project developed to estimate employee salaries based on various factors such as Age, Gender, Education Level, Job Title, and Years of Experience.

This project demonstrates the complete Machine Learning lifecycle, including data preprocessing, exploratory data analysis, feature engineering, model building, model evaluation, and deployment using Streamlit.

The final solution is an interactive web application where users can enter employee information and instantly receive a salary prediction generated by the trained Machine Learning model.



Problem Statement

Organizations often need to determine employee salaries based on multiple factors such as experience, education, and job role.

Manual salary estimation can be inconsistent and time-consuming. Therefore, a Machine Learning-based solution is developed to predict employee salaries accurately using historical employee data.



Project Objectives

- ✓ Analyze employee salary data
- ✓ Perform data cleaning and preprocessing
- ✓ Understand salary influencing factors
- ✓ Build a Machine Learning prediction model
- ✓ Evaluate model performance
- ✓ Save the trained model for future use

✓ Develop an interactive Streamlit application

✓ Generate real-time salary predictions

Dataset Description

The dataset contains employee-related information and corresponding salary values.

Input Features

- ◆ Age
- ◆ Gender
- ◆ Education Level
- ◆ Job Title
- ◆ Years of Experience

Target Variable

 Salary

The Salary column serves as the dependent variable that the model predicts.

Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning.

Steps Performed

- ✓ Dataset Inspection
- ✓ Data Type Verification
- ✓ Feature Selection
- ✓ Label Encoding
- ✓ Train-Test Split

Label Encoding

Categorical variables were converted into numerical values using Label Encoding.

Encoded Columns:

- Gender
- Education Level
- Job Title

This transformation enabled the Machine Learning algorithm to process categorical information effectively.



Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns, relationships, and trends within the dataset.

Analysis Performed



Salary Distribution Analysis



Age Distribution Analysis



Experience Analysis



Education Level Analysis



Correlation Analysis

Key Insights



Years of Experience is the strongest factor affecting salary.



Employees with higher education levels generally earn higher salaries.



Senior job roles are associated with significantly higher salaries.



Salary tends to increase with professional experience.

These findings helped identify the most influential features for model development.



Feature Selection

The dataset was divided into independent variables and the target variable.

Independent Variables (X)

- Age
- Gender
- Education Level
- Job Title
- Years of Experience

Target Variable (Y)

- Salary

These variables were used to train the Machine Learning model.

Train-Test Split

The dataset was split into:

 80% Training Data

 20% Testing Data

The training dataset was used to train the model, while the testing dataset was used to evaluate its performance on unseen data.

Machine Learning Model

Algorithm Used

- ♦ Linear Regression

Linear Regression was selected because it is a simple yet effective algorithm for predicting continuous numerical values such as salaries.

Why Linear Regression?

- ✓ Easy to understand
- ✓ Fast training process
- ✓ Suitable for regression problems

- ✓ Ideal for beginner Machine Learning projects
- ✓ Provides interpretable results

Model Training

The model was trained using the training dataset to learn relationships between employee attributes and salary values.



Salary Prediction

After successful training, the model generated salary predictions using employee information.

The trained model learns historical patterns and estimates salaries for new employee records entered through the Streamlit application.



Model Evaluation

The model's performance was measured using standard regression metrics.

Evaluation Results



Mean Absolute Error (MAE)

22,937



Root Mean Squared Error (RMSE)

30,076



R² Score

0.67

Interpretation

The R² Score of approximately 0.67 indicates that the model explains about 67% of the variation in employee salaries.

This demonstrates acceptable predictive performance for a basic Linear Regression model.

Model Saving

The trained model was saved using the Pickle library.

Saved Model File

 salary_model.pkl

Benefits

- ✓ Avoids retraining the model
- ✓ Faster deployment
- ✓ Easy integration with applications
- ✓ Reusable for future predictions

Streamlit Application Development

An interactive Streamlit application was developed to make salary prediction accessible through a user-friendly interface.

User Inputs

-  Age
-  Gender
-  Education Level
-  Job Title Code
-  Years of Experience

Application Workflow

User Input



Data Encoding



Load Saved Model



Generate Prediction



Display Predicted Salary

The application provides instant salary predictions based on the entered information.

Technologies Used

Programming Language

 Python

Libraries

 Pandas

 NumPy

 Matplotlib

 Seaborn

 Scikit-Learn

 Pickle

 Streamlit

Development Tools

 Jupyter Notebook

 VS Code

 GitHub

 Streamlit Cloud



Project Workflow

 Data Collection



 Data Cleaning



 Exploratory Data Analysis



 Label Encoding



 Feature Selection



 Train-Test Split



 Linear Regression Model Training



 Model Evaluation



 Model Saving



 Streamlit Deployment



 Real-Time Salary Prediction

Key Learnings

Through this project, I gained practical experience in:

- ✓ Data Cleaning and Preparation
- ✓ Exploratory Data Analysis
- ✓ Feature Engineering
- ✓ Label Encoding
- ✓ Linear Regression Modeling
- ✓ Model Evaluation
- ✓ Pickle Model Serialization
- ✓ Streamlit Development
- ✓ End-to-End Machine Learning Workflow

Conclusion

This project successfully developed and deployed a Machine Learning model capable of predicting employee salaries based on employee characteristics such as age, gender, education level, job title, and years of experience.

The analysis identified Years of Experience and Education Level as key factors influencing salary. A Linear Regression model was trained and evaluated, achieving satisfactory predictive performance.

The trained model was saved using Pickle and integrated into a Streamlit web application, enabling users to obtain real-time salary predictions through a simple and interactive interface.

Overall, this project provided valuable hands-on experience in building a complete Machine Learning solution, covering every stage from data preprocessing and analysis to deployment and practical implementation.